

UNIT-II

Energy Storing Devices

LP2: Irreversible cell: Primary battery (Alkaline and Zinc-air battery)-Construction, working and applications

Primary batteries:

Primary batteries are not easily rechargeable, and consequently are discharged then disposed of. Many of these are “dry cells” – cells in which the electrolyte is not a liquid but a paste or similar.

The cell electrochemical reactions are not easily reversible and cell is operated until the active components in the electrodes are exhausted. Generally primary batteries have a higher capacity and initial voltage than rechargeable batteries.

Advantages:

- Inexpensive
- Convenient
- Lightweight
- Good shelf life
- High Energy density at low/moderate discharges

Disadvantages:

- Can only be used once
- Leads to large amount of waste batteries to be recycled
- Batteries put into landfill sites have severe environmental impact
- Life cycle energy efficiency < 2 %

The table below demonstrates the properties of various primary batteries:

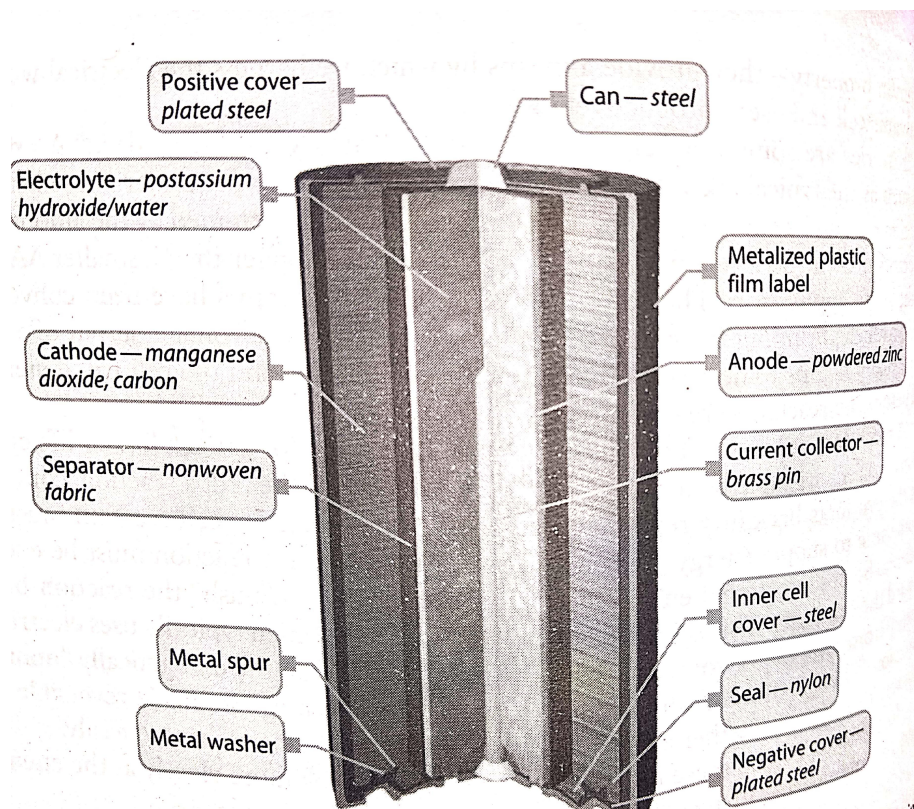
System	Nominal Cell Voltage (V)	Capacity (Wh/kg)	Advantages	Disadvantages	Applications
<u>Carbon/Zinc</u>	1.50	65	Lowest cost; variety of shapes and sizes	Low energy density; poor low-temperature performance	Torches; radios; electronic toys and games
Mg/MnO₂	1.60	105	Higher capacity than C/Zn; good shelf life	High gassing on discharge; delayed voltage	Military and aircraft receiver-transmitters
<u>Zn/Alk/MnO₂</u>	1.50	95	Higher capacity than C/Zn; good low-temperature performance	Moderate cost	Personal stereos; calculators; radio; TV
Zn/HgO	1.35	105	High Energy density; flat discharge; stable voltage	Expensive; energy density only moderate	Hearing aids; pacemakers; photography; military sensors/detectors
Cd/HgO	0.90	45	Good high and low-temperature performance; good shelf life	Expensive; low energy density	
<u>Zn/Ag₂O</u>	1.50	130	High Energy density, good high rate performance	Expensive (but cost effective)	Watches; photography; missiles; Larger space applications
Zn/Air	1.50	290	High Energy density; long shelf life	Dependent on environment; limited power output	Watches; hearing aids; railway signals; electric fences
Li/SOCl₂	3.60	300	High Energy density; long shelf life	Only low to moderate rate applications	Memory devices; standby electrical power devices
Li/SO₂	3.00	280	High energy density; best low-temperature performance; long shelf life	High-cost pressurized system	Military and special industrial needs
Li/MnO₂	3.00	200	High energy density; good low-temperature performance; cost effective	Small in size, only low-drain applications	Electrical medical devices; memory circuits; fusing

Alkaline battery

Alkaline battery is a type of primary battery, which cannot be recharged. The electrochemical reaction that happens in the battery is not reversible. The alkaline battery gets its name because it has an alkaline electrolyte of potassium hydroxide.

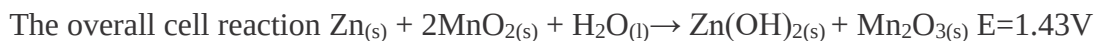
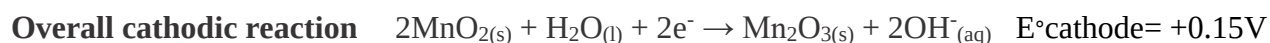
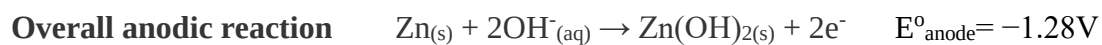
Construction

- An alkaline battery consists of a outer cylindrical can made of steel and it holds the cathode connection
- Compressed paste of manganese dioxide (MnO_2) with carbon powder is the positive electrode (cathode)
- Carbon powder is added for increasing conductance property
- Each cathode has a hollow centre and lined with a separator made up of cellulose and synthetic polymer that prevents the contact of the electrode material
- The separator has the conducting property, so it conduct ions and also remains stable with the alkaline potassium hydroxide electrolyte solution
- A mixture of zinc powder with potassium hydroxide electrolyte which is made in to a gel is the negative electrode (anode)



Working or Discharging:

- In an alkaline battery, Zinc act as negative electrode (anode) manganese dioxide act as a positive electrode (cathode)
- Potassium hydroxide doesn't take part in the reaction
- The alkaline electrolyte remains same as there is an equal amount of OH^- ion consumed and produced.
- At the anode zinc powder around the central metal rod is oxidised to form Zn^{2+} , which reacts immediately with OH^- ions in the electrolyte to form zinc hydroxide.
- At the cathode Manganese dioxide is reduced from Mn^{4+} oxidation state to Mn^{3+} oxidation state:



Advantages:

- 50-100% more energy than carbon zinc
- Low self-discharge (10 year shelf life)
- Good for low current ($< 400\text{mA}$)
- long-life use

Usage of **alkaline batteries** for low-drain **applications** offers commendable performance with very low replacement rate. Low-drain **applications** include flashlights, portable radios, alarm clocks, remote controls, toys and the like. Low-drain **applications** dominate the **alkaline battery** market.

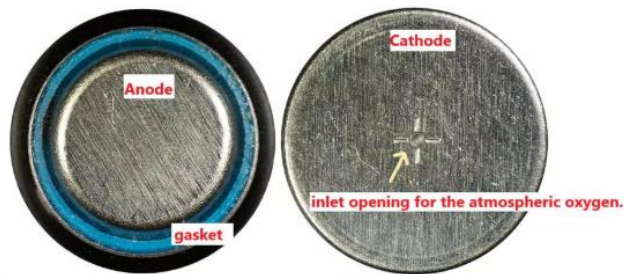
Zinc-Air battery

The production of electrochemical energy of battery is due to the use of oxygen from atm. Diffused oxygen acts as cathode. Air cathode catalytically promotes the reaction of oxygen with electrolyte (KOH). The outer metal acts as the anode of the battery. The output of the Zinc-air battery is 1.65V.



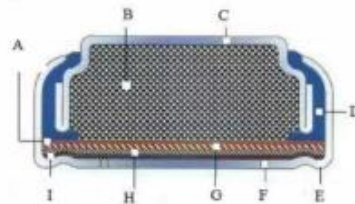
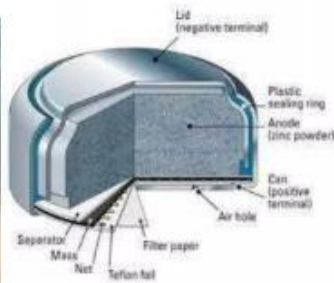
- Zinc air batteries and zinc-air fuel cells are metal air batteries which are powered by oxidizing zinc with oxygen from the air
- Zinc-air batteries have high energy density
- Zinc-air batteries range from button cell, film cameras, electrical vehicle propulsion

ZINC AIR BATTERIES



CONSTRUCTION & WORKING

- **PRIMARY (NON-RECHARGEABLE)**
Button cell type and 1Ah type
- **SECONDARY (RECHARGEABLE)**



- A: Separator, B: zinc powder anode, C: anode can, D: insulator gasket, E: cathode can, F: air hole, G: cathode catalyst and current collector, H: air distribution layer, I: Semi permeable membrane

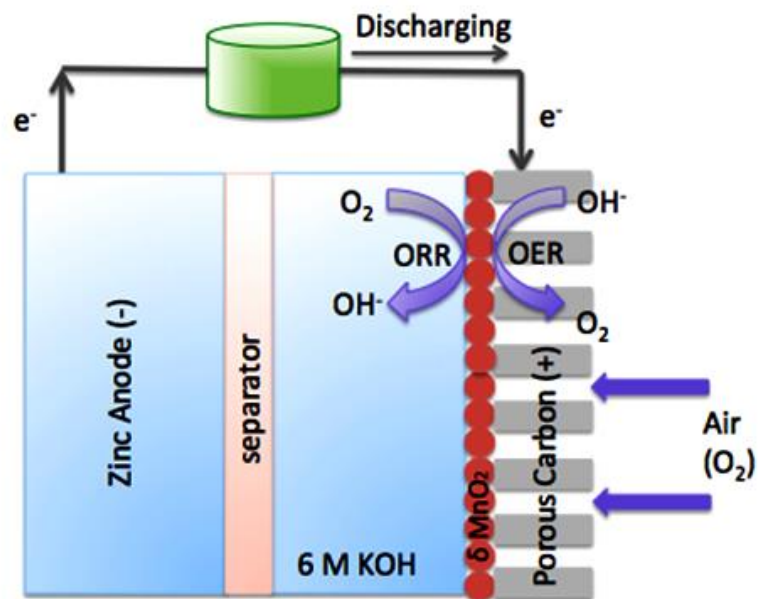
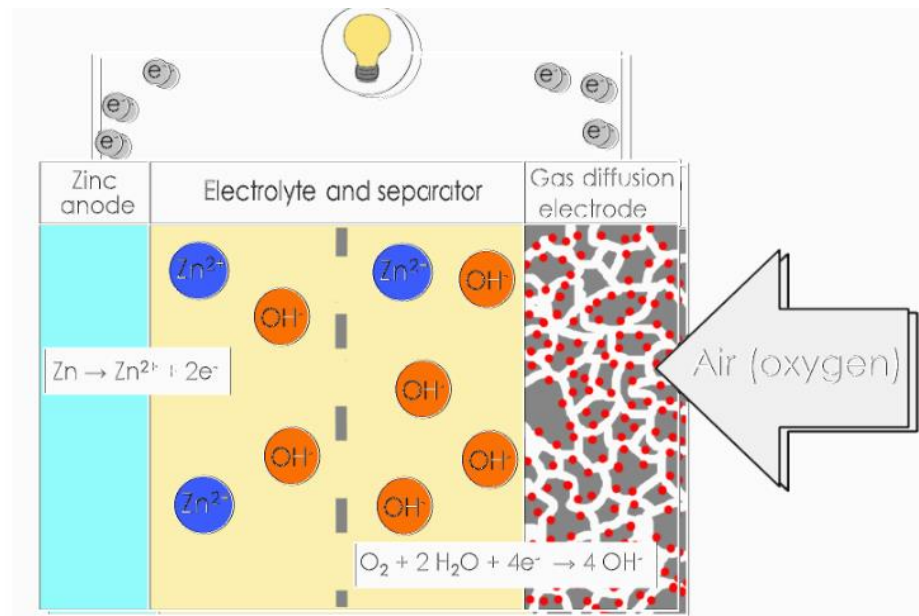
In Zinc-air battery the oxygen from the air reacts at the cathode and forms hydroxyl ions which migrate into the zinc paste and form zincate ($\text{Zn}(\text{OH})_4^{2-}$), releasing electrons to travel to the cathode.

The zincate decays into zinc oxide and water return to the electrolyte.

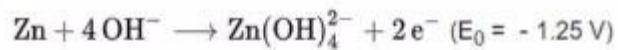
The water and hydroxyl from the anode and recycled at the cathode, so water is not consumed

The reaction produce a theoretical 1.65 volts, but this is reduced to 1.35-1.4 V in available cells.

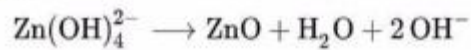
Working



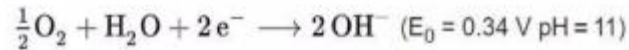
Anode:



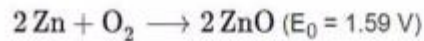
Fluid:



Cathode:



Overall



Advantages

- Easy to carry
- High energy density
- Competitive with lithium ion
- Inexpensive
- Flat discharge voltage
- Safety- do not require volatile materials, so zinc –air batteries are not prone to catching fire like lithium battery
- Excellent self life, with a self discharge rate of only 2%
- Available in range of button and coin cell sizes
- Environmental benefits
- Have high volumetric energy density compared to most primary batteries

Disadvantages

- Sensitive to extreme temperature and humid conditions
- Carbon dioxide from the air forms carbonate which reduces conductivity
- High self discharge after seal is broken
- After activation chemicals tend to dry out and the batteries have to be used quickly
- Zin-air battery must be larger to satisfy high current needs
- Have flooding potential
- Limited output